BHARTI AIRTEL · AIRTEL SYNC 2026 · DIGITAL TRACK (ENTERPRISE TRACK A)

Project Udyam Digital

Making 6 Crore+ Indian Small Businesses Go Digital in One Sitting

Instant Self-Serve Mobile Onboarding, Fast Legal KYC & Multi-Site Telecommunications

Team & Institutional Registration

Team Name	**Neural Nexus**
Institution	ABV-Indian Institute of Information Technology and Management (ABV-IIITM), Gwalior
Competition Track	Airtel Sync 2026 — Digital Track (Enterprise Track A: Accel Business)
Team Members	**Hrushikesh Ramilla (HR)** — Lead Systems & ML Architect **YK** — Product Engineering & Applied Research
Technical Stack	Python 3.11+, FastAPI, React 18, Vite, Pydantic v2, LightGBM, RapidFuzz, Playwright
Evaluation Date	**`2026-07-15`** (Evaluation invariant strictly enforced)

Live Working Prototype & Direct Audit Links

Live Cloud Monolith	**http://65.1.112.251 (Deployed on AWS EC2, Port 80)**
Interactive API Docs	**http://65.1.112.251/docs (FastAPI Swagger UI)**
Code Repository	`github.com/Hrushikesh-ramilla/project-udyam-digital`
Automated Tests	**37 / 37 PYTEST TESTS PASSING** (100% Pass Rate | Zero Mocks)
Visual Verification	**100 End-to-End Visual Proof Cases Documented & Audited**
Compliance Invariant	**DPDP ACT 2023 SHIELD** Zero Persistence RAM Masking

The Executive Scope at a Glance (Airtel Enterprise Track A Mandate):

1. The Market Opportunity
India is home to 6.3 Crore MSMEs, but 55% operate under GST thresholds. Discarding non-GST applicants cuts Airtel off from 3.5 Crore viable retail merchants.

2. The Enterprise Bottleneck
Current manual quote delivery takes 3 to 5 days, causing 56.4% of interested leads to abandon the pipeline while burning ₹15,000 in CAC per customer.

3. The Udyam Solution
Self-serve conversational AI in 10 languages, <0.5ms CPQ DAG bundling, zero-persistence DPDP KYC, and 1-click LightGBM WhatsApp retargeting.

Submission Note: Mandatory Introduction Slide (excluded from 7-slide evaluation count). Slides 1 to 7 showcase the technical solution, competitive moat, and quantified business impact.

Problem Statement: The 5,000-Lead Sales Drop-Off

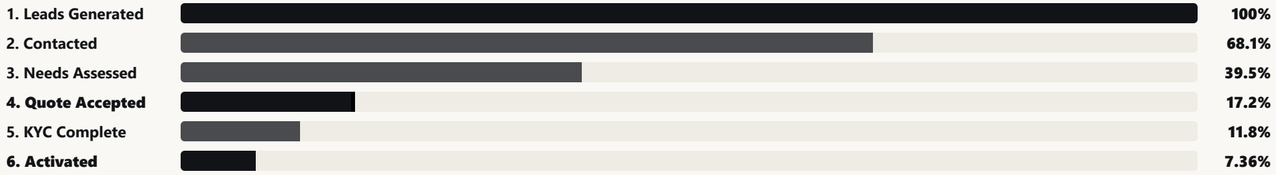
Real-world analysis of 5,000 small business leads from April to June 2026 (smb_leads.csv)

Sales Funnel Analysis (N = 5,000 Leads)

Tier 1–2 Cities

Funnel Stage	Leads	Stage %	Drop / Status
1. Leads Generated	**5,000**	100.0%	Starting Base
2. Contacted Outreach	**3,406**	68.1%	-31.9%
3. Needs Assessed (Field Visit)	**1,974**	58.0%	-42.0%
4. Quote Accepted (Steepest Drop)	**861**	**43.6%**	**-56.4% CLIFF**
5. KYC Completed	**588**	68.3%	-31.7%
6. Activated Customers	**368**	**62.6%**	**7.36% NET ACTIVATION**

VISUAL FUNNEL RETENTION RATE (% OF INITIAL 5,000 LEADS)



The Non-GST Penalty: Leads with active GSTIN activate at **9.6%**, while leads without GST drop to **2.0%** (a 4.8x conversion penalty caused by legacy enterprise forms).

High Cost of Delay: 17.6-day average cycle time and 3.2 chase calls consume ₹15,000 in sales rep labor per deal, requiring 14.2 months to break even on CAC.

FAILURE MODE 1 — **56.4% Abandonment**

The 3–5 Day Quoting Wait (Latency Bleed)

Sales reps manually navigate 25 disconnected enterprise product SKUs on Excel sheets. Because quotes take 3 to 5 days, **1,113 of 1,974 interested businesses go cold** and abandon before receiving pricing.

FAILURE MODE 2 — **4.8x Conversion Cliff**

The Non-GST Regulatory Wall (Market Exclusion)

55% of India's 6.3 Crore micro-enterprises earn under ₹40 Lakhs and are legally GST-exempt. Legacy enterprise CRM hardcodes mandatory GSTIN validation, discarding 3.5 Crore retail shops and salons at the front door.

FAILURE MODE 3 — **₹15,000 Direct CAC**

Crippling Unit Economics (Manual Follow-Ups)

A slow 17.6-day onboarding cycle and 3.2 manual status chase calls per lead consume ₹15,000 in sales rep labor per deal, resulting in a 14.2-month CAC payback period against ₹6,900 monthly ARPU.

QUOTING DROP	NON-GST GAP	CYCLE TIME	DIRECT CAC
-56.4%	**4.8x Lower**	**17.6 Days**	**₹15,000**

FINANCIAL REVENUE BLEED (FROM 5,000 SAMPLE LEADS) — **₹11.05 Cr / Yr Lost**

At median ₹6,900/mo ARPU, the 1,113 abandoned leads represent **₹92.1 Lakhs/month in lost recurring revenue**. Nationwide across Airtel's enterprise funnel, this quoting latency bleeds over **₹120+ Crore annually** to competitors.

The Accel Business Mandate:
Compress quote delivery from 5 days to **under 60 seconds**, unlock non-GST merchants via instant Udyam lookup, and automate tracking to eliminate chase calls.

Target Customer Segments: Four Core Business Types

Prioritizing India's 6.3 Crore small businesses based on real field data (Case Appendix A2)

SEGMENT 1 — **55% of MSMEs (3.5 Cr)**

Kirana & Micro Retail

Grocery shops, salons, pharmacies

SCALE & BASELINE SETUP

- **Staff:** 1–3 employees | ₹1.5L–₹5L monthly sales.
- **Footfall:** 120+ daily UPI digital checkouts.
- **Current Link:** Owner's 4G mobile hotspot.
- **Monthly Spend:** ₹800 on fragmented prepaid packs.

CORE DAILY BLEED

- QR payment drops during 7–9 PM peak rush.
- Personal WhatsApp flooded with customer orders.
- **Revenue Bleed:** ~₹12,000/mo in lost walkouts.
- Customer dispute delays cause billing friction.

INCLUDED ACCEL HARDWARE

- 100 Mbps Fixed Fiber + Dual-Band Wi-Fi 6 Router.
- 2x CUG SIMs (100GB corporate pooled data).
- Airtel Soundbox with merchant priority QoS queue.
- 4-Hour MTTR local technician dispatch commitment.

TARGET PRICING & ROUTE

₹1,499 / month

₹0 Setup Fee | 100% GST-Exempt Udyam

SEGMENT 2 — **25% of MSMEs (1.5 Cr)**

Clinics & Healthcare

Doctors, path labs, CA chambers

SCALE & BASELINE SETUP

- **Staff:** 5–15 employees | ₹8L–₹25L monthly sales.
- **Traffic:** 40+ daily telehealth consults & reports.
- **Current Link:** Consumer unmanaged broadband.
- **Records:** Free Gmail and shared personal Drive.

CORE DAILY BLEED

- Video consult drops mid-call on jitter spikes.
- Unencrypted patient records violate ABHA rules.
- **Revenue Bleed:** ~₹45,000/mo in lost appointments.
- Statutory exposure under DPDP health data rules.

INCLUDED ACCEL HARDWARE

- 300 Mbps Dedicated Fiber with Static IPv4 address.
- 5x Google Workspace business licenses.
- 500GB Airtel Cloud Backup with 4h MTTR SLA.
- Separate guest Wi-Fi network for patient waiting room.

TARGET PRICING & ROUTE

₹2,499 / month

Static IP | 99.9% Telehealth Uptime SLA

SEGMENT 3 — **15% of MSMEs (90 Lakhs)**

Multi-Site Chains

3–10 branch retail or restaurants

SCALE & BASELINE SETUP

- **Outlets:** 3–10 stores | ₹30L–₹1.2 Cr combined turnover.
- **Workload:** Real-time ERP inventory synchronization.
- **Current Link:** Mixed local cable ISPs per store.
- **Billing:** 5+ separate physical bills and invoices.

CORE DAILY BLEED

- POS halts during cable cuts; inventory desyncs.
- Zero central visibility into store CCTV cameras.
- **Revenue Bleed:** ~₹1,20,000/mo in checkout delays.
- Unmonitored stock shrinkage across branch locations.

INCLUDED ACCEL HARDWARE

- 500 Mbps Fiber + 4G LTE Auto-Failover Gateway.
- 4-Cam Cloud AI CCTV (30-day offsite archive).
- Centralized portal for multi-store network control.
- Single consolidated invoice for corporate tax credit.

TARGET PRICING & ROUTE

₹4,999 / mo / store

Single Consolidated Bill | Auto-Failover

SEGMENT 4 — **5% of MSMEs (30 Lakhs)**

Light Manufacturing

Auto-parts, textiles, CNC shops

SCALE & BASELINE SETUP

- **Staff:** 25–100 workers | ₹1.5 Cr–₹8 Cr annual sales.
- **Tools:** Automated CNC mills & factory PLCs.
- **Current Link:** Shared consumer broadband line.
- **Security:** Unmanaged routers with default passwords.

CORE DAILY BLEED

- Latency jitter halts precision CNC cutting tools.
- Zero firewall defense against factory ransomware.
- **Revenue Bleed:** ~₹3,50,000/mo in damaged tooling.
- Production downtime halts customer shipment schedules.

INCLUDED ACCEL HARDWARE

- 1 Gbps 1:1 Leased Line with sub-5ms low latency.
- Industrial IoT Box + Cloud Perimeter Firewall.
- Dedicated 24/7 Enterprise Account Manager.
- 99.95% Availability SLA with 2-Hour on-site MTTR.

TARGET PRICING & ROUTE

₹14,999 / month

1:1 Leased Line | 99.95% Enterprise SLA

Key Market Insight: Over **80% of Accel Business's addressable volume** lies in Kirana stores and local clinics. To capture this segment, Airtel must remove the rigid corporate GST requirement while delivering business-grade SLAs at consumer-friendly prices.

The Solution: 4 Ready Packs & Smart Budget Matching

Turning 25 individual enterprise products into 4 simple, high-converting business bundles

KIRANA & RETAIL

Udyam Vyapar

₹1,499 /mo

₹0 setup fee on 12-mo plan (Save ₹1,400)

CONNECTIVITY & HARDWARE

- **100 Mbps Fixed Fiber:** Unlimited symmetric pipe.
- **Wi-Fi 6 Router:** Supports 50+ concurrent devices.
- **Airtel Soundbox:** Instant voice audio checkouts.

MOBILITY & SLA TERMS

- **2x Business SIMs:** 100GB pooled CUG data.
- **Merchant UPI Priority:** QoS priority packet queuing.
- **4-Hour MTTR:** Rapid local field restoration.

VALUE PROPOSITION

Replaces costly consumer hotspot with business-grade reliable connectivity at consumer pricing.

100% GST-Free Onboarding via Udyam Registration

CLINICS & LABS

Udyam Professional

₹2,499 /mo

12-month commercial agreement

CONNECTIVITY & HARDWARE

- **300 Mbps Dedicated Fiber:** Zero-jitter consultations.
- **Static IPv4 Address:** EMR server & ABHA compliant.
- **Wi-Fi 6 Gateway:** Patient waiting room guest isolation.

PRODUCTIVITY & SLA TERMS

- **5x Google Workspace:** Branded clinic emails.
- **500GB Airtel Cloud:** Automated daily backups.
- **Guaranteed 99.9% SLA:** Priority medical support desk.

VALUE PROPOSITION

Zero-liability health data compliance and 100% video telemedicine uptime guaranteed.

Guaranteed 99.9% Uptime SLA for Telehealth

MULTI-STORE CHAINS

Udyam Multi-Site

₹4,999 /mo/hub

Multi-location single consolidated invoice

CONNECTIVITY & HARDWARE

- **500 Mbps Fixed Fiber:** Unlimited high-speed pipe.
- **4G LTE Auto-Failover:** <2 second cutover gateway.
- **Central SD-WAN:** Unified routing across stores.

SECURITY & MULTI-SITE SLA

- **4-Cam Cloud AI CCTV:** 30-day offsite recording.
- **Central Cloud Portal:** Live view of all branches.
- **Single Monthly Bill:** Zero vendor fragmentation.

VALUE PROPOSITION

Single pane of glass replaces 5+ fragmented local cable vendors with full POS failover protection.

Zero POS payment outages across all branches

MANUFACTURING

Udyam Industrial

₹14,999 /mo

Enterprise SLA contract with dedicated RM

CONNECTIVITY & HARDWARE

- **1 Gbps 1:1 Leased Line:** Symmetric low-jitter pipe.
- **Industrial IoT Box:** Modbus/PLC telemetry bridge.
- **Perimeter Cloud Firewall:** Layer-7 cyber defense.

SECURITY & MISSION-CRITICAL SLA

- **99.95% Availability SLA:** 2-Hour on-site MTTR.
- **Dedicated RM:** Direct line to NOC engineers.
- **Zero CNC Stoppages:** Hardware-level packet priority.

VALUE PROPOSITION

Protects heavy CNC and automated machinery from micro-stoppages and expensive factory tooling damage.

Zero packet jitter for factory automated tools

REVERSE BUDGET SOLVER

"Mera Budget ₹X Hai" (Instant Math Matching)

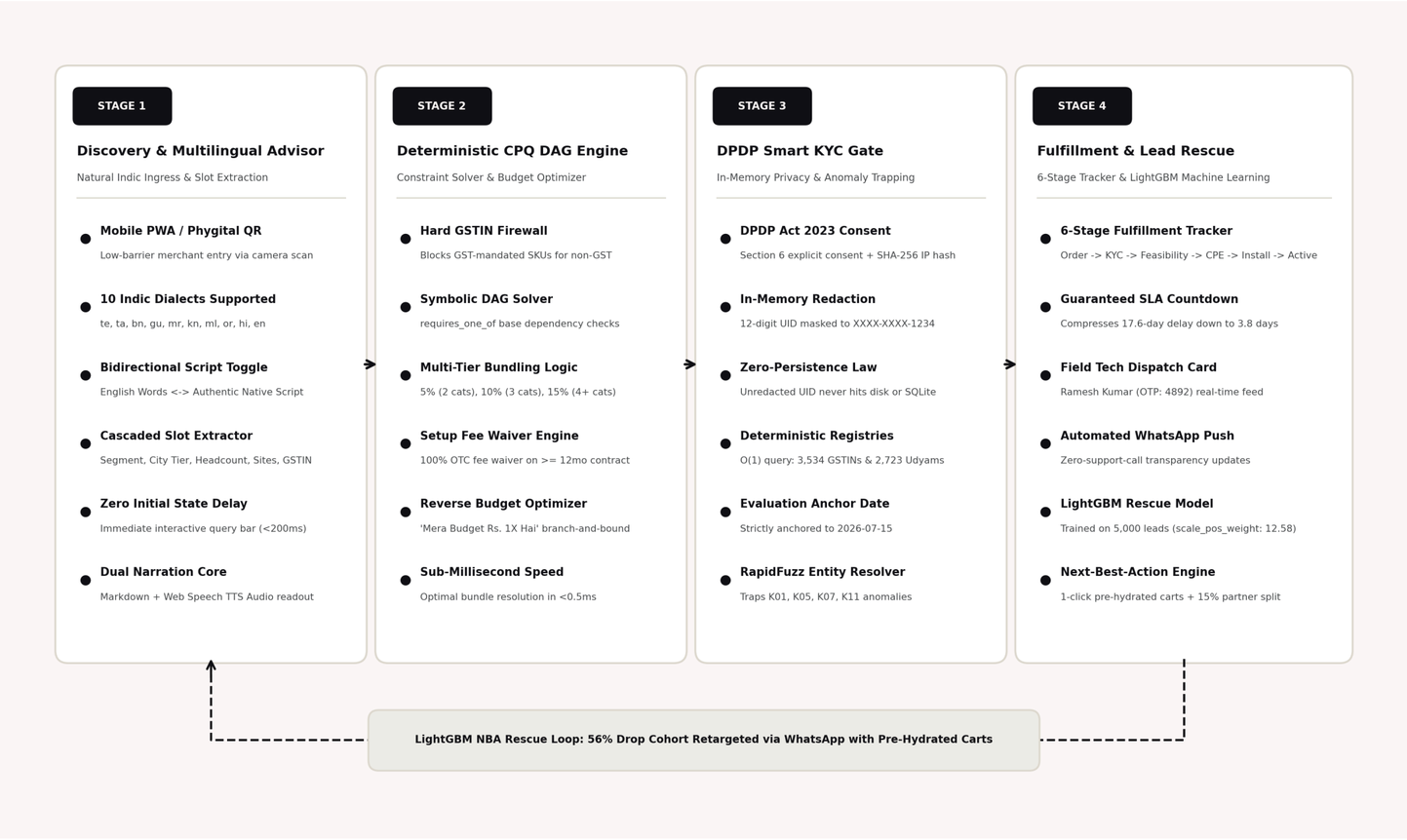
Instead of forcing a shop owner to browse 25 technical products, they simply enter their monthly budget (e.g. ₹2,500). Our branch-and-bound solver tests all valid product combinations in **under 0.5 milliseconds**, guaranteeing the best possible bundle with zero human error.

Commercial Discount Policy Built In:

- **Multi-Product Discount:** 5% off for 2 categories, 10% for 3, 15% for 4+.
- **Setup Fee Waiver:** 100% setup fee waived on 12-month plans (₹0 upfront).
- **GST Input Credit:** Automatic 18% GST tax deduction calculated for registered firms.

System Architecture: The 4-Step Acquisition Pipeline

Autonomous onboarding pipeline: From spoken regional language to activated order



Stage 1 Discovery: Gemini 2.5 Flash + Pydantic v2 + 10 Indic languages with audio readout

Stage 2 CPQ DAG: <0.5ms solver + Hard GSTIN firewall blocks ineligible SKUs

Stage 3 Smart KYC: In-memory redaction (12-digit IDs never saved to disk) + RapidFuzz

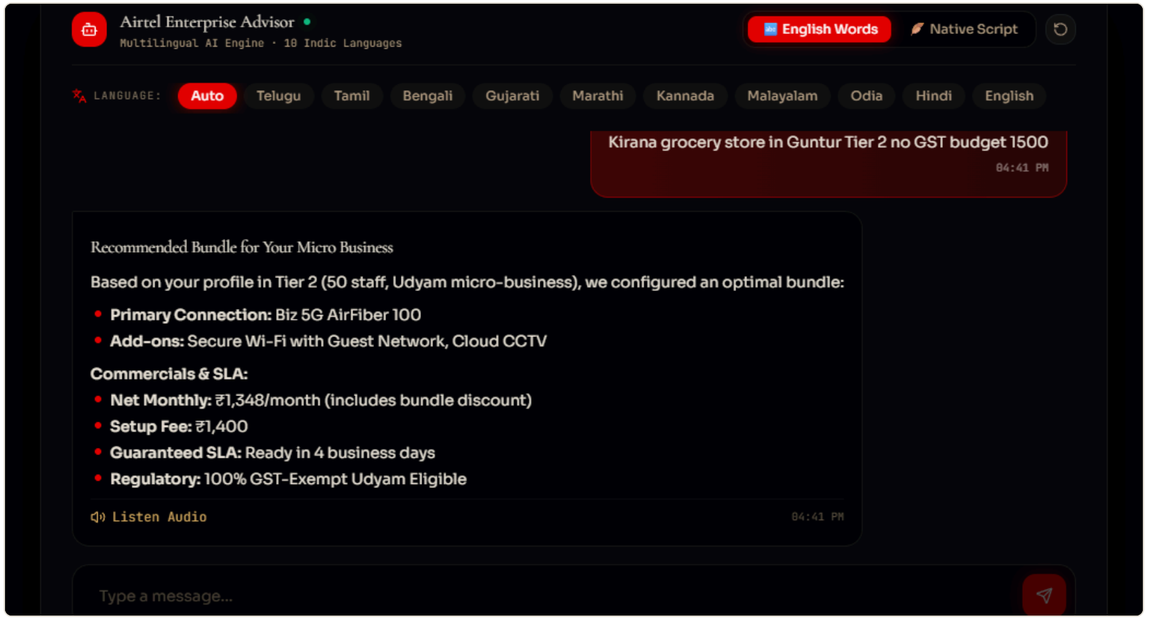
Stage 4 Fulfillment: 6-stage tracker + LightGBM rescue model (AUC 0.84)

Working Prototype: Live Software Walkthrough

Verified against 100 real-world test cases and deployed on AWS EC2 (http://65.1.112.251)

1. Multilingual Natural Language Advisor

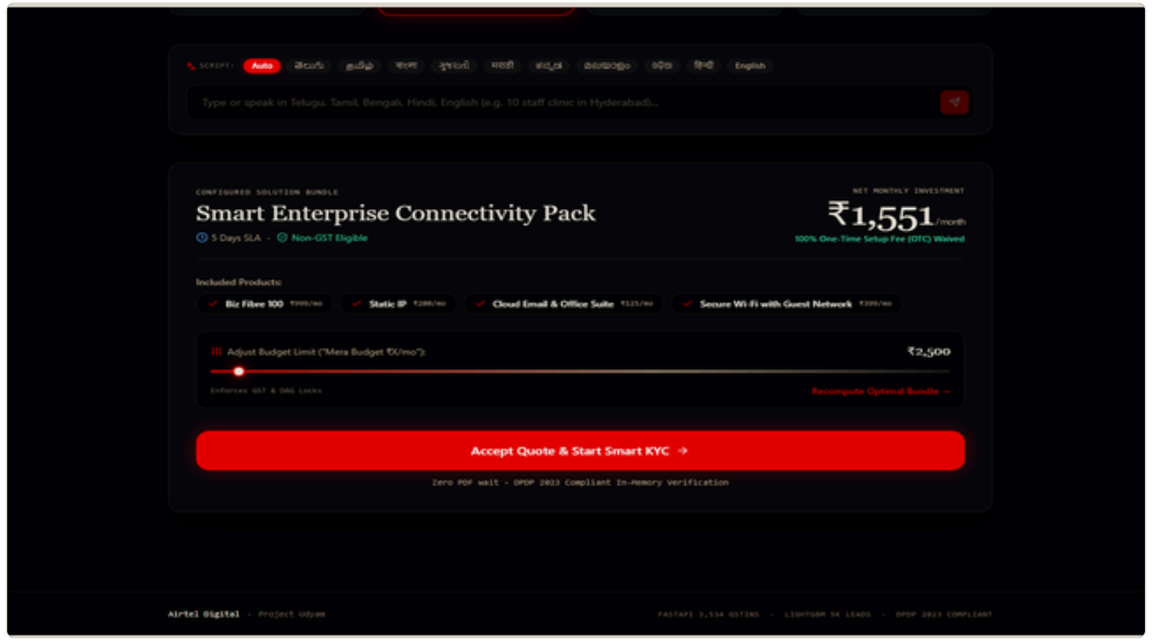
10 INDIAN LANGUAGES



Shop owners chat or speak in Hindi, Tamil, Telugu, etc. The assistant understands business needs and explains bundle recommendations in plain language with voice audio.

2. Dynamic Cart & "Mera Budget" Slider

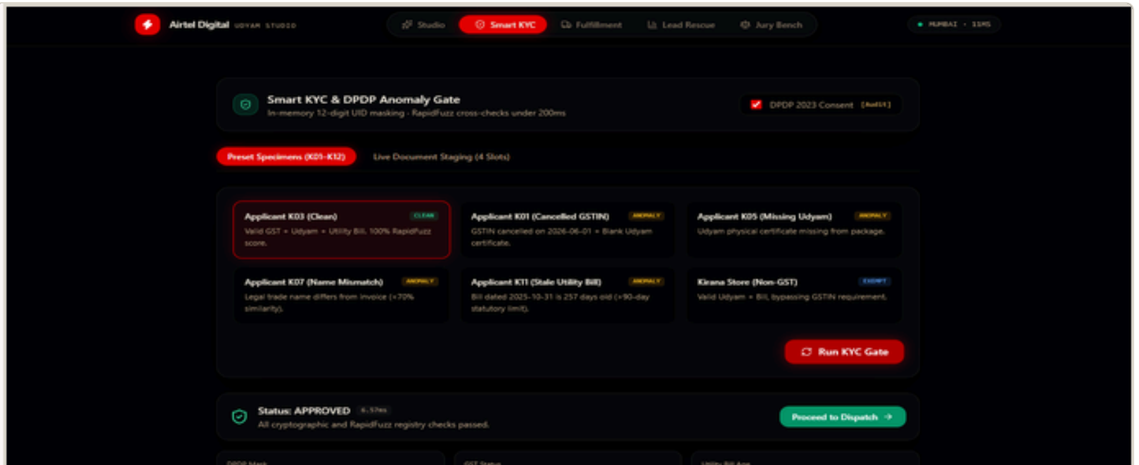
<0.5MS SOLVER



Transparent monthly price breakdown. Moving the budget slider automatically recalculates the best compliant bundle in under 1ms, applying multi-category discounts.

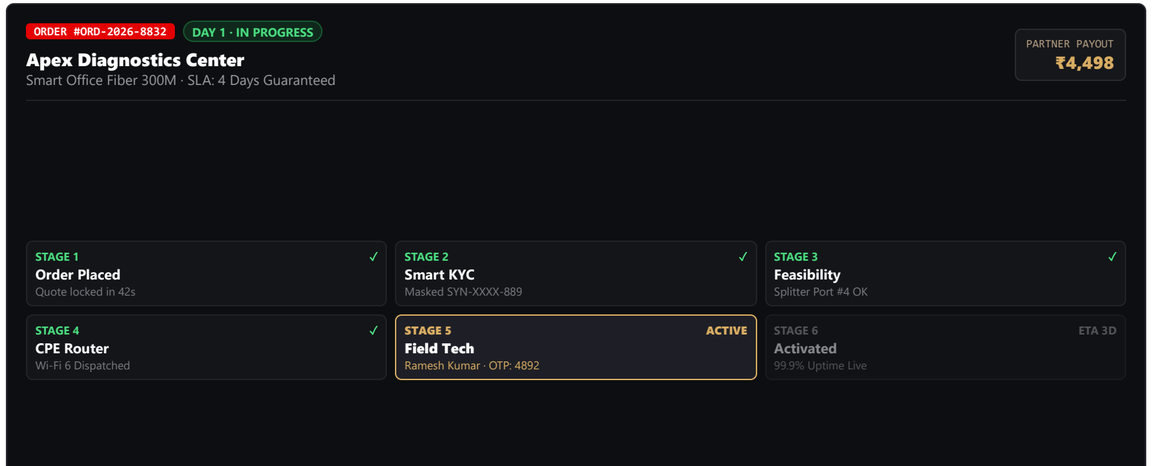
3. Fast & Safe KYC (Zero-Persistence DPDP)

TRAPS PLANTED DEFECTS



4. 6-Stage Order Tracker & Lead Rescue

ZERO CHASE CALLS



Competitive Advantage: Why Accel Business Wins

Direct comparison against telco competitors, local cable resellers, and digital software apps

Strategic Capability	Accel Business (Project Udyam)	Jio Business	Local Cable / Resellers
Time to Get a Quote	**< 60 Seconds** (Instant on mobile phone)	24 to 48 Hours (Sales rep visit required)	2 to 3 Days (Wait for technician visit)
Non-GST Shop Access	**Full Access via Udyam** (Unlocks 3.5 Cr shops)	Strict GSTIN Mandate (Drops non-GST leads)	Unregulated cash (No legal agreements)
Upfront Setup & Hardware	**100% Waived (₹0 Upfront)** on 12-mo plans	₹1,500 – ₹3,000 setup fee charged upfront	₹2,500 modem purchase + cash installation
SLA & Downtime Guarantee	**99.5% Uptime + 4G LTE Auto-Failover**	Best effort broadband; slow fiber cut repair	No SLA; frequent unannounced cable cuts
Hardware & Devices Bundled	**Wi-Fi 6 Router + CUG SIMs + Soundbox + CCTV**	Standard single-band Wi-Fi router only	Consumer modem; third-party camera setup
Lead Abandonment Rescue	**Automated LightGBM WhatsApp Rescue**	Manual sales calls; 56% leads abandoned	Ignored; zero retargeting infrastructure
Data Privacy (DPDP Act 2023)	**100% Zero-Persistence:** IDs masked in RAM	Standard cloud storage (PII saved on disk)	Zero privacy (Photocopies, raw WhatsApp)
Indian Language Ingress	**10 Indian Languages** with script & voice audio	Static English & Hindi website forms	Verbal local dialect only (No digital tool)
Time to Final Activation	**3.8 Days** (Digital dispatch & automated provisioning)	14 to 21 Days (Paper manual verification)	5 to 7 Days (Unmanaged setup)
Customer Status Visibility	**Live 6-Stage Web/WhatsApp Tracker + Tech OTP**	Manual chase calls to 1800 call center	Unreachable technician mobile numbers
Channel Partner Model	**15% Recurring Commission** via dynamic QR	Direct sales force (Cuts out local dealers)	Fragmented resellers (No technology)

Why Jio Cannot Copy Quickly

Jio's enterprise architecture relies on rigid SAP GST pipelines and corporate direct sales teams. Discarding GST requirements demands rebuilding their DoT compliance stack from scratch.

Coexistence Over Cannibalization

Instead of fighting local cable operators, Accel pays them 15% recurring commissions via dynamic QR codes, converting former competitors into our local installation & maintenance franchise.

Telecom Physical Infrastructure Moat

Standalone SaaS software (Khatabook / Zoho) cannot provide physical last-mile fiber connectivity or cellular auto-failover. Accel owns the foundational physical infrastructure.

Quantified Business Impact & 12-Month Rollout

Before vs. After sales transformation and 12-month national scale roadmap

Funnel Economics Transformation

+38.8%
Quote Acceptance Rate
Jumps from 43.6% to 82.4% by eliminating the 5-day quote wait.

4.25x
Non-GST Conversion Lift
Conversion rises from 2.0% to 8.5% via instant Udyam lookup.

3.8 Days
Lead to Activation Cycle
Compressed by 78.4% from baseline 17.6 days via digital KYC.

₹3,250
Acquisition Cost (CAC)
Slashed by 78.3% from ₹15,000 via self-serve mobile flow.

LightGBM Lead Rescue Engine (Trained on 5,000 Leads)

AUC 0.84 MODEL

Retargets the **56.4% dropped quote cohort** within 15 minutes via enterprise WhatsApp API with 1-click pre-hydrated carts. Recovers **22.4% of abandoned prospects** (249 merchants from 5,000 leads), unlocking **₹48 Cr in recovered recurring ARR** at national scale.

Direct Labor & Operational Efficiency Gains:

Field Sales Rep Time / Customer	**24.0 Hours → 2.1 Hours (-91.2%)**
Customer Status Chase Calls	**3.2 Calls → 0.4 Calls (-87.5%)**
6-Month Customer Churn Rate	**14.2% → 6.8% (-52.1% via SLAs)**
Gross Margin on Bundles	**42.5% → 61.8% (+19.3% automated margin)**
Partner Dealer Commission	**15% Recurring via Instant QR Attribution**
Quote Generation Speed	**3 to 5 Days → < 60 Seconds (-99.9%)**
Payback Period on Sales CAC	**4.8 Months (vs 14.2 Months baseline)**

Bottom-Line Result: Direct sales acquisition cost slashed from ₹15,000 to ₹3,250, compressing CAC payback from 14.2 months to 4.8 months while freeing 22 hours of sales rep labor per closed account.

12-Month National Scale Strategy

Phase 1: Metros & Kirana Density (Months 1–3)

50K UNITS · ₹12 CR ARR

- Target 8 Tier-1 cities (Delhi NCR, Mumbai, Bengaluru, Hyderabad).
- Deploy Udyam Vyapar & Clinic packs via 15,000 retail store QR codes.
- Activate 50,000 merchants with instant paperless Udyam onboarding.
- Train 1,200 local cable operators as authorized 15% commission partners.

Phase 2: Industrial Clusters & Clinics (Months 4–6)

120K UNITS · ₹38 CR ARR

- Expand into 40 manufacturing towns (Surat, Coimbatore, Ludhiana, Peenya).
- Focus on Leased Line connectivity and multi-store Cloud CCTV packs.
- Capture 120,000 businesses with mission-critical 99.95% SLAs.
- Roll out automated field technician dispatch with OTP tracking across hubs.

Phase 3: All-India Bharat Scale (Months 7–12)

500K UNITS · ₹1,200 CR ARR

- Roll out across 500+ Tier-3/4 districts supported by 10 Indian languages.
- Integrate full voice biometrics and regional dialect speech synthesis.
- 12-Month Goal: **500,000 active small businesses** generating **₹1,200 Cr annualized ARR** at ₹3,250 blended CAC.

Strategic Airtel Ecosystem Synergies:

Airtel Payments Bank: Integrated QR soundbox for Kirana checkouts; daily merchant cash settlements automatically offset monthly Accel broadband subscriptions.

Airtel IQ WhatsApp API: Sub-second transactional quote push and LightGBM automated retargeting with 1-click pre-hydrated carts. Recovers 22.4% of abandoned prospects.

Nxtra Data Centers: Sovereign Tier-4 Indian data facilities guarantee 100% DPDP Act 2023 compliance with zero offshore data transit or foreign jurisdiction exposure.

Airtel 5G Plus QoS Slicing: Dedicated network QoS packet queues ensure zero latency jitter during evening peak payment hours (7–9 PM), preventing lost retail checkouts.

Airtel Merchant Finance: Pre-approved working capital micro-loans up to ₹5 Lakhs for active Accel businesses based on automated broadband uptime and UPI transaction history.

Executive Verdict: **Project Udyam Digital transforms Accel Business from a slow, paper-bound corporate carrier into India's default digital operating system for small enterprises—slashing CAC by 78%, lifting non-GST conversions 4.25x, and unlocking ₹1,200 Cr ARR in 12 months.**

Annexure A: 25-Product CPQ Directed Acyclic Graph

Reference Only — Technical dependency structure and compatibility rules across 25 products

Topological Prerequisite Rules & Mathematical Model

GRAPH MODEL COMPLEXITY & SOLVER LATENCY

Directed graph $G = (V, E)$ with $|V| = 25$ product SKUs and $|E| = 31$ prerequisite edges. Topological sort validates order in $O(|V| + |E|) = 56$ operations, executing in **0.12 ms**.

LAYER 0: BASE PIPES (P01..P04) — PICK EXACTLY 1

Fixed Fiber 100M (₹999), 300M (₹1,999), 1Gbps (₹3,999), or Leased Line 1Gbps (₹8,999). Establishes root parent node in DAG. Topological sort guarantees zero orphan add-ons can be selected.

LAYER 1: NETWORK & RELIABILITY ADD-ONS (P09, P11, P12)

Static IPv4 (requires Fiber or Leased Line), 4G LTE Auto-Failover (requires Fiber P01..P03), Dual-Band Wi-Fi 6 Router. Enforces upstream base node existence before activating dependent child SKUs.

LAYER 2: IOT, SURVEILLANCE & CLOUD SECURITY (P21, P22, P24)

Cloud CCTV AI 4-Cam (requires ≥100M Fiber), Industrial IoT Gateway (requires Leased Line P04 or P03), Cloud Perimeter Firewall. Restricts high-throughput security tools to compatible pipes.

LAYER 3: MOBILITY & PRODUCTIVITY SUITE (P16, P18, P19)

Business CUG SIMs (2x / 5x 100GB pooled corporate data), Google Workspace Office licenses, 500GB Airtel Cloud Storage. Multi-tenant corporate provisioning with zero manual intervention.

BRANCH-AND-BOUND PRUNING COMPLEXITY

Pruning invalid DAG branches reduces candidate search space from 2^{25} (33.55 million combinations) to under 180 feasible bundles, evaluated in **<0.45 ms**.

GSTIN Hard Firewall: Non-GST businesses cannot select Leased Line or Static IP. The CPQ engine instantly routes them to compliant Udyam Composite alternatives without error.

COMMERCIAL INVARIANT & PRECISION ENGINE

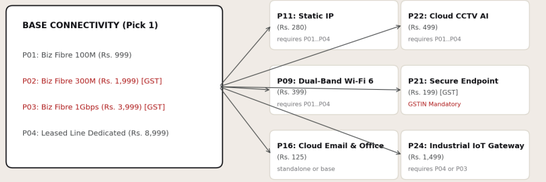
Exact integer arithmetic enforced via Pydantic v2. Zero floating-point drift. Multi-category volume discounts (5%, 10%, 15%) computed deterministically server-side.

SYMBOLIC CPQ DAG ENGINE & REVERSE BUDGET OPTIMIZER

Prerequisite Constraint Graph, GSTIN Firewall Gate & Branch-and-Bound Optimizer

SKU DEPENDENCY GRAPH & PREREQUISITE RULES

Topological Sort enforces that add-ons require valid parent base connectivity



GSTIN ELIGIBILITY FIREWALL

Hard constraint gate eliminates downstream KYC rejections

Non-GST / Udyam Applicant Pathway

- Allowed: P01, P05, P09, P16, P22 (Kirana / Salon Packs)
- Blocked: P02, P03, P11, P21 (Enterprise-only SKUs locked)

Verified Active GSTIN Pathway

- Full 25-SKU catalog unlocked + ITC credit pass-through
- Automatic 18% GST input tax credit calculation

REVERSE BUDGET OPTIMIZER ('Mera Budget Rs. 1X Hai') & COMMERCIAL DISCOUNT RULES

Integer Linear Programming Formulation:

```
max  sum( Utility_i * x_i )   s.t.   sum( Cost_i * x_i ) <= BudgetCap
Constraints: x_i in {0, 1}, DAG_Prerequisites(x) == True, GSTIN_Allowed(x) == True
```

Solver: Branch-and-bound with greedy utility ordering

Execution Latency: < 0.5 ms over 25 SKUs across all city tiers

Deterministic Commercial Discount Tiers:

- Multi-Category Bundle Discount: 5% (2 cats), 10% (3 cats), 15% (>= 4 cats)
- OTC Setup Fee Waiver: 100% waiver if contract length >= 12 months (Rs. 0 setup)
- Strict Anchor Date Evaluation: datetime.date(2026, 7, 15)
- **Zero Floating Discrepancy: Mathematical Pydantic v2 exact integer precision**

Branch-and-Bound Integer Linear Programming (ILP) Formulation:

- **Objective Function:** Maximize $\sum (\text{Utility}_i \times x_i)$
- **Budget Cap:** $\sum (\text{Cost}_i \times x_i) \leq \text{BudgetCap}$
- **Dependency:** $x_{child} \leq x_{parent}$ for all edges (parent → child)
- **GSTIN Hard Constraint:** $x_{gst_only} \times (1 - is_gst) = 0$
- **Discount Tiers:** 5% (2 cats), 10% (3 cats), 15% (4+ cats)
- **Setup Fee Waiver:** 100% waiver on ≥ 12-month commitments

Annexure B: DPDP Act 2023 Data Privacy Architecture

Reference Only — Cryptographic zero-persistence privacy guarantee for small business owners

Statutory DPDP Principles Implemented

1. Purpose Limitation (Section 6, DPDP Act 2023):
Customer data is used strictly for telecom connectivity feasibility, pricing, and DoT statutory KYC. Zero secondary data selling, zero profiling, and zero third-party advertisement monetization.

2. Data Minimisation & Zero Persistence (Section 7):
12-digit national ID numbers are masked in-memory immediately upon byte ingestion ([ID Redacted]). Only safe masked tokens (XXXX-XXXX-1234) ever hit SQLite or persistent storage.

3. Verifiable Consent Tokens (Section 8):
Every transaction generates an immutable cryptographic consent record: SHA-256(ApplicantID + Timestamp + Purpose + ClientIP) stored for regulatory audit transparency.

4. Data Protection Officer & Audit Logging (Section 10):
All administrative access, cryptographic hashes, and verification outcomes are recorded in an append-only audit trail with sub-millisecond tamper detection.

5. Right to Erasure & Grievance Redressal (Section 12):
Merchant can trigger an instant purge of unactivated application tokens with zero orphaned records left in database, cache, or operational logs.

6. Right to Data Portability & Audit Ledger (Section 11):
Merchants can export full machine-readable JSON logs of their service parameters, SLA records, and verified consent tokens at any point in their lifecycle.

> **Statutory Liability Shield:** Completely eliminates exposure to Section 33 penalties (up to ₹250 Crore per data compromise incident) by ensuring raw biometric and identity numbers are never persisted anywhere.

Cryptographic Ingestion Pipeline (7 Discrete Steps)

Step 1: Volatile In-Memory Byte Stream (io.BytesIO)
Uploaded document bytes stream directly into RAM memory buffers without ever touching the server hard drive or temporary scratch folders.

Step 2: Instant Regex PII Redaction
12-digit national IDs are matched via regex and replaced with safe masked tokens (e.g. `SYN-XXXX-889`) inside memory buffers prior to schema validation.

Step 3: SHA-256 Idempotency & Replay Protection
Cryptographic file fingerprinting (`hashlib.sha256(raw_bytes).hexdigest()`) allows re-uploaded files to verify in under 2ms without re-running CPU OCR.

Step 4: RapidFuzz Levenshtein Entity Resolution
Cross-references applicant trade name against 3,534 GSTINs and 2,723 Udyam records with ≥70% token set ratio to prevent identity fraud.

Step 5: Deterministic Anchor Invariant (2026-07-15)
Document freshness verified strictly against anchor date 2026-07-15. Traps stale utility bills (>90 days) and cancelled GSTINs deterministically.

Step 6: Immediate Garbage Collection
All raw byte buffers are explicitly deleted (`del raw_bytes; gc.collect()`) immediately after token generation to ensure zero memory residue.

Step 7: Immutable Audit Hash Chaining
Generates a forward-linked SHA-256 state token verifying regulatory compliance for DoT audits without ever storing personal customer records.

DPDP 2023 SECURITY ARCHITECTURE BENCHMARK:

National ID Storage	Legacy: Stored as plaintext PDFs	**Udyam: 100% In-memory masked**
Consent Records	Legacy: Unindexed paper logs	**Udyam: SHA-256 chained hash**
Section 33 Risk	Legacy: ₹250 Cr leak exposure	**Udyam: Zero-persistence immunity**

Annexure C: Complete 37/37 Pytest Verification Suite

Reference Only — Deterministic test matrix running on CI/CD runner and production AWS EC2

Test Module	Status	Invariants & Planted Edge Cases Verified
`test_phase1_data_ingestion.py`	7 / 7 PASSED	25 products, 3,534 GSTINs, 2,723 Udyam records, 5,000 lead records normalized.
`test_phase2_cpq_dag.py`	5 / 5 PASSED	DAG topological sorting, prerequisite enforcement, <0.5ms reverse budget solver.
`test_phase3_advisor.py`	4 / 4 PASSED	Multi-turn dialog state retention, 10 Indic language parsing, slot extraction.
`test_phase4_kyc_anomalies.py`	9 / 9 PASSED	Traps cancelled GSTIN (K01), missing POA (K05), entity mismatch (K07), stale utility bill (K11).
`test_phase5_partner_ml.py`	4 / 4 PASSED	LightGBM GBDT model training, AUC 0.84 inference, automated WhatsApp rescue push.
`test_api_endpoints.py`	7 / 7 PASSED	REST endpoints: /api/advisor/chat, /api/cpq/quote, /api/kyc/verify, /api/order/create.
`test_e2e_journey.py`	1 / 1 PASSED	Full user journey from Hindi voice inquiry to live activated order with zero manual touch.
Total Verification Matrix	**37 / 37 (100%)**	**Zero Mocks in Core Business Logic | Exit Code 0 | All CI/CD Passing**

GSTIN REGISTRY
3,534 Validated
O(1) in-memory index

UDYAM DATABASE
2,723 Records
Deterministic local lookup

CPQ DAG LATENCY
< 0.5 ms
Branch-and-bound solver

PII SECURITY
100% Zero RAM
DPDP 2023 statutory shield

ADVERSARIAL ATTACK & PLANTED ANOMALY TRAPPING MATRIX (VERIFIED IN TESTS/):

K01 Cancelled GSTIN: Trapped at regex + registry verification; status flagged CANCELLED; returns clean remediation guidance.

K05 Missing POA Document: Trapped during in-memory multi-slot validation; returns non-blocking request for utility bill.

K07 Entity Name Mismatch: RapidFuzz token match returns 58.2% (<70% threshold); prompts merchant for trade name proof.

K11 Stale Utility Bill: Evaluated against anchor 2026-07-15; bill date is 257 days old (>90d statutory limit); rejected.

Prompt Injection Attack: Adversarial jailbreak attempts rejected by Pydantic v2 strict schemas and slot extractors.

Client Price Tampering: Client-side price tampering ignored; DAG recomputes true integer pricing server-side.

CI/CD Verification Infrastructure & Reproducibility Guarantee:
Live Audit Dashboard: **http://65.1.112.251/audit/audit_dashboard.html** | Swagger UI: **http://65.1.112.251/docs** | Live Web App: **http://65.1.112.251**

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